

Convergence Investigation Report — case-b

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Investigation Identities

Original: **caseB_alpha**
Candidate: **caseB_beta**

Linkage Strength-of-Evidence

95.7/100 — STRONG

This score represents strength of evidence, not probability of guilt or identity certainty.

Signal Breakdown

Signal	Similarity	Contribution	Status	Evidence ID
temporal	0.839	+17.87	SUPPORTING	E-caseB_alpha-caseB_beta-temporal
account	1.000	+15.98	SUPPORTING	E-caseB_alpha-caseB_beta-account
infrastructure	0.950	+12.92	SUPPORTING	E-caseB_alpha-caseB_beta-infrastructure
activity	1.000	+19.30	SUPPORTING	E-caseB_alpha-caseB_beta-activity
interaction_network	0.994	+16.42	SUPPORTING	E-caseB_alpha-caseB_beta-interaction_network
linguistic	0.995	+13.23	SUPPORTING	E-caseB_alpha-caseB_beta-linguistic

Supporting Evidence

temporal: {'matching_activity_hours': [9, 10, 11, 20, 21], 'matching_activity_windows': ['09:00–12:00', '20:00–22:00'], 'observation_period_days': 14, 'similarity': 0.839}

account: {'creation_gap_days': 0.0, 'posting_interval_diff_minutes': 0.0, 'creation_decay_days': 60, 'posting_decay_minutes': 120, 'similarity': 1.0}

infrastructure: {'identity_a_infra': 'infra_DARKNET_CLUSTER_7', 'identity_b_infra': 'infra_DARKNET_CLUSTER_7', 'matched_components': ['tls', 'service', 'certificate', 'server', 'asn'], 'match_ratio': 1.0, 'match_type': 'exact', 'identity_a_components': {'tls': 'tls_DARK', 'service': 'svc_DARK', 'certificate': 'cert_DARK', 'server': 'srv_DARK', 'asn': 'asn_DARK'}, 'identity_b_components': {'tls': 'tls_DARK', 'service': 'svc_DARK', 'certificate': 'cert_DARK', 'server': 'srv_DARK', 'asn': 'asn_DARK'}, 'similarity': 0.95, 'ps26151_infrastructure_evidence': {'ssl_certificates': ['cert_DARK'], 'certificate_domain_links': [{'certificate': 'cert_DARK', 'domain': 'shadowfox.example'}], 'clearnet_domains': ['shadowfox.example'], 'service_banners': ['svc_DARK'], 'exposed_server_status': ['server-status: enabled'], 'descriptor_inconsistencies': ['descriptor_version_overlap']}, 'observations': ['Shared SSL certificate identifier(s).', 'SSL certificate tied to the same clearnet-domain relationship.', 'Shared clearnet domain identifier(s).', 'Shared service/default banner(s).', 'Shared exposed server-status observation(s).', 'Shared descriptor inconsistency marker(s).'], 'note': 'Infrastructure score uses TLS/service/certificate/server/ASN component overlap; PS-specific clearnet/certificate/server-status/descriptor objects are explicitly represented as corroborating evidence.'}

activity: {'identity_a_actions': {'search': 6, 'withdraw': 2, 'message': 6, 'browse': 4, 'upload': 3, 'post': 1, 'login': 2}, 'identity_b_actions': {'search': 6, 'withdraw': 2, 'message': 6, 'browse': 4, 'upload': 3, 'post': 1, 'login': 2}, 'strongest_matching_behaviors': ['message', 'search', 'browse'], 'similarity': 1.0}

interaction_network: {'identity_a_interaction_vector': [9.802, 8.518, 10.89, 13.631, 9.342], 'identity_b_interaction_vector': [9.436, 10.425, 10.441, 12.793, 7.837], 'shared_counterparties': ['cp_13', 'cp_14', 'cp_17', 'cp_8'], 'shared_counterparty_count': 4, 'similarity': 0.994, 'note': 'Behavioral interaction patterns are attribution evidence; privacy/anonymity tools are excluded from this signal.'}

linguistic: {'similarity': 0.995, 'features': ['sentence length', 'punctuation pattern', 'function-word usage', 'vocabulary patterns', 'n-gram distribution'], 'representation': 'Synthetic stylometric representation', 'note': 'Prototype stylometry is synthetic; the similarity computation itself is cosine similarity.'}

Contradicting Evidence

None identified.

Insufficient Evidence

None identified.

Threat-Actor Intelligence Layer

Shared objects: **14**

Handles: ['ShadowFox']

PGP keys: ['PGP_SHADOWFOX_7821']

Wallets: ['WALLET_SHADOWFOX_91']

Trust links: ['TRUST_MARKET_7']

Marketplaces: ['market_alpha']

Cleartext domains: ['shadowfox.example']

SSL certificates: ['cert_DARK']

Certificate ↔ domain links: [{'certificate': 'cert_DARK', 'domain': 'shadowfox.example'}]

Service/default banners: ['svc_DARK']

Exposed server-status: ['server-status: enabled']

Descriptor inconsistencies: ['descriptor_version_overlap']

Hidden-service identifiers: ['shadowfox_onion_91']

Persona history overlap: ['ShadowFox:v1', 'ShadowFox:v2-rebrand']

This layer is corroborating context and is not a seventh score signal.

Privacy / Anonymity Context — NOT Attribution Evidence

Original: {'tor': True, 'vpn': True, 'cryptocurrency': True, 'encrypted_messaging': False}

Candidate: {'tor': False, 'vpn': False, 'cryptocurrency': False, 'encrypted_messaging': True}

These indicators do not contribute to the attribution score.

Methodology

Temporal cosine similarity; configurable account decay; infrastructure component overlap; activity cosine similarity; interaction-network similarity; synthetic stylometry; weighted evidence fusion; separate contradiction penalty; confidence bands.

Evidence Integrity

E-caseB_alpha — SHA-256: 9fca07d332fb4474732266f342dd03afb0b1a4323953a5c04dcb1e62e52c1641

E-caseB_beta — SHA-256: 4f5e55c27210fb53c43e1b682755df628fdae5e3b1442f3783e292bf9d538bbd

Limitations

Controlled synthetic evaluation only. It does not establish real-world de-anonymization performance.

Decision-Support Notice: This report does not establish real-world identity or guilt.